Proving Lines Parallel — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see "Your turn."

Name: _____ Date: _____

☞ *Video: Run it backwards*

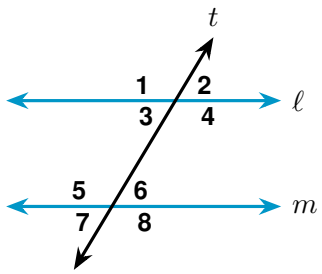
Key Idea

Last lesson: lines parallel ⇒ angles congruent.

This lesson: angles congruent ⇒ **lines parallel**. These are called **converses**.

☞ *Video: The converses*

Key Idea



Corresponding angles congruent ⇒ $\ell \parallel m$

Alt. interior **or** alt. exterior angles **congruent** ⇒ $\ell \parallel m$

Same-side interior angles **supplementary** ⇒ $\ell \parallel m$

Two lines parallel to the **same** line are parallel.

☞ *Video: Prove it*

Watch Me — Worked Example

$m\angle 4 = 72°$, $m\angle 5 = 72°$. Is $\ell \parallel m$?

1. Name the pair: **alternate interior**
2. Same measure, so $\angle 4 \cong \angle 5$
3. So $\ell \parallel m$ by the **Alternate Interior Angles Converse**

☞ *Video: Find x so the lines are parallel*

Watch Me — Worked Example

Same-side interior: $(4x + 6)°$ and $(5x - 15)°$.

Need a **sum** of $180°$: $9x - 9 = 180 \Rightarrow x =$ **21**

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☞ *Video: Watch out*

Common Mistake

Never use "they **look parallel**." Name the angle pair, show it is congruent (or supplementary), then name the **converse**.

☞ *Video: Your turn*

Your Turn

$m\angle 1 = 115°,\ m\angle 8 = 115°$. Is $\ell \parallel m$? Which converse?

$\angle 1$ and $\angle 8$ are **alternate exterior** angles, and they are congruent.

$\ell \parallel m$? **yes** by the **Alternate Exterior Angles Converse**

Notice

Remember: Congruent corresponding or alternate angles $\Rightarrow$ parallel. Supplementary same-side interior angles $\Rightarrow$ parallel. Name the pair, show the angle fact, name the converse.

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